

## LLM: Data

`.gitignore` – Repository ga push qilyotkanda, `repo` ga chiqmaydigan `file` yokida `folder`larni bekitib quyadigan `muhim file`.

Massive `modelni weightlari`, `datasetlar`, `token` va `API` kabi juda sirli va `private` `malumotlarni` `.gitignore` ga solamiz

## Data Sources

| Dataset      | Domain       | URL                  | Use Case                                          |
|--------------|--------------|----------------------|---------------------------------------------------|
| Common Crawl | Web          | <a href="#">LINK</a> | Raw web data for data collection & cleaning       |
| FineWeb      | Web Text     | <a href="#">LINK</a> | High-quality text for general SLM pretraining     |
| FineWeb-Edu  | Education    | <a href="#">LINK</a> | Educational text for an education SLM             |
| The Pile     | Multi-Domain | <a href="#">LINK</a> | Books, papers, code, web for general SLM training |
| RedPajama    | Multi-Domain | <a href="#">LINK</a> | Open data for general-purpose SLMs                |
| Dolma        | Multi-Domain | <a href="#">LINK</a> | Mixed corpus for pretraining experiments          |
| The Stack v2 | Code         | <a href="#">LINK</a> | Source code for a Code SLM                        |
| arXiv        | Academic     | <a href="#">LINK</a> | Research papers for Scientific SLMs               |

## Best usage of Data

| Goal           | Dataset      | Why?                         |
|----------------|--------------|------------------------------|
| General SLM    | FineWeb      | Clean, high-quality web text |
| Education SLM  | FineWeb-Edu  | Educational content          |
| Code SLM       | The Stack v2 | Programming code             |
| Scientific SLM | arXiv        | Academic & STEM content      |

Biz qilmoqchi bulgan `maqsadimizga qarab`, tugri va maqsadga `'mos'` `data source` dan foydalansak buladi.

## Quality Dimension

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katta hajmli dataset hardoimla zur quality ga ega bulmaydi. Bu holatni biz Supervised-machine learning dan boshlab kurib kelimiz.

Hatto filterlanib va deduplication qilgandan keyinham quyidagi featurelarga ahimiyat berish Kerak:

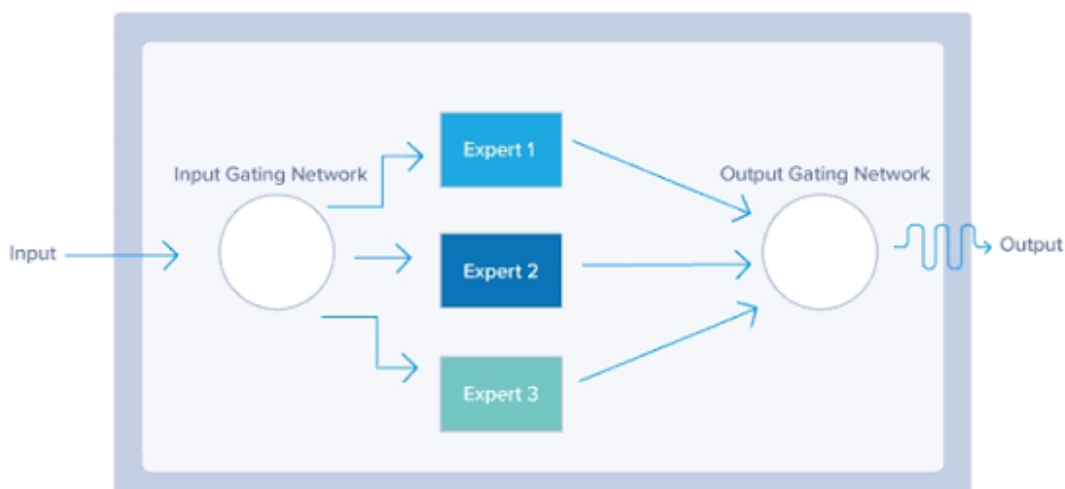
1. **Relevance**: Dataset modelning maqsadiga mos bulishi kerak. A spam, casino, reklamalar bulmasi kerak
2. **Quality**: Text grammatik tomondan tugri, tushunarli va foydali bulishi kerak. Shunchaki internetdagi har qanday text training uchun yaxshi degani emas
3. **Diversity**: Dataset turli xil source lardan iborat bulishi kerak: web, kitoblar, science, code, history, news va boshqalar. Bu modelni bir yunalishga boglanib qolishidan saqlaydi
4. **Accuracy & Factual Quality**: Internet malumotlari xatolar kup bulgani uchun matnning tugriligi va haqiqiyligini tekshirish
5. **Safety & Toxicity**: Conflictli, zararli, xavfli yoki noqonuniy contentlarni aniqlash va Filterlash
6. **Duplication**: Malumotlar tuplamida ortiqcha takrorlanishlarning oldini olish uchun bir xil malumotlarni topish va uchirish.
7. **PII / Sensitivity**: Private malumotlar (name, phone-num, e-mail, etc.) modelga uqitilmasligi uchun ularni anonimlashtirish va himoya qilish.

## Data Mixture

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LLM da faqat bitta dataset bilan training qilinmaydi. Yani, bir nechta data sourcelardan kerakli qismlarini olib natijada yaxshi mix bulingan data orqali training qilamiz

### MoE (Mixture of Experts LLM)



Bir nechta sourcelardan faqat kerakli qismlarini olgan holda final data chiqadi. Yuqoridagi modelni featurelardan biri **diversion** degani bor edi. Yani, modelimiz kup yunalishdagi sourcelardan iborat bulish kerak edi.

Va **foiz** (%) berishniham **stragety buyicha berishi** kerak. Shunchaki random bermasdan.

## License

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Open-source data bulsaham, **100% hohlaganday ishlatsa bumed**  
Shunga, noqonuniy bulib qolmasligi uchun, **dataset license check** qilish kerak:

- qanaqa license?
- datasetni egasi kim?
- nimalarga ruhsat bor?
- business holda foydalanish mumkinmi?
- biroga junatsam buladimi?
- qanaqa tusqinliklar bor?

License turlari:

| License            | Commercial? | Modify? | Redistribute? |
|--------------------|-------------|---------|---------------|
| <b>CC0</b>         | Yes         | Yes     | Yes           |
| <b>CC BY</b>       | Yes         | Yes     | Yes           |
| <b>CC BY-SA</b>    | Yes         | Yes     | Yes           |
| <b>CC BY-NC</b>    | No          | Yes     | Depends       |
| <b>CC BY-NC-SA</b> | No          | Yes     | Depends       |
| <b>CC BY-ND</b>    | Yes         | Depends | Depends       |
| <b>CC BY-NC-ND</b> | No          | Depends | Depends       |
| <b>ODC-By</b>      | Yes         | Yes     | Yes           |
| <b>ODbL</b>        | Yes         | Yes     | Depends       |
| <b>MIT</b>         | Yes         | Yes     | Yes           |
| <b>Apache-2.0</b>  | Yes         | Yes     | Yes           |
| <b>BSD-3</b>       | Yes         | Yes     | Yes           |
| <b>GPL-3.0</b>     | Yes         | Yes     | Depends       |